Multisource Heterogeneous Transfer Learning via Feature Augmentation for Ship Classification in SAR Imagery

Haitao Lang, *Member, IEEE*, Guang'an Yang, Chunnan Li, and Jianwen Xu

***Abstract*— Improving ship classification performance in synthetic aperture radar (SAR) imagery by the methods based on transfer learning (TL) is a newly emerging research topic and has great potential. The existing studies merely address the problem of TL from a single source domain (SD) [automatic identification system (AIS) or optical remote sensing (ORS)] to the target domain (TD) (SAR) based on the homogeneous TL (HoTL), which requires all domains to be represented by homogeneous features with the same dimensions. Our work takes a step forward and attempts to address a more meaningful and challenging problem that transfers knowledge from multiple SDs for the purpose of exploring and exploiting complementarity cross SDs to assist ship classification in the TD. To this end, our study develops a multisource heterogeneous transfer learning (MS-HeTL) method that liberates the restriction of utilizing the exact same features for all domains, allows each domain represented by a more appropriate feature, and, thus, improves the ship classification performance even further. Specifically, we first propose multisource heterogeneous feature augmentation (MS-HFA) to effectively solve the challenges brought by feature heterogeneity and fully exploit the complementarity cross domains. Then, a support vector machine (SVM) classification framework is specific-designed to be incorporated with the augmented feature representations in the common space to conduct knowledge transfer cross domains. Extensive experiments on two benchmark datasets, HR-SAR and FUSAR, show that the proposed method outperforms the existing methods and demonstrates its effectiveness and advantages. All datasets and source codes are available at https://github.com/BUCT-RS-ML/MS-HeTL-via-MS-HFA.**

***Index Terms*— Heterogeneous feature augmentation (HFA), heterogeneous transfer learning (HeTL), multisource HeTL (MS-HeTL), multisource HFA (MS-HFA), ship classification, synthetic aperture radar (SAR).**

I. Introduction

SHIP classification based on high-resolution synthetic aperture radar (SAR) imagery has attracted more and more attention in related research communities, including remote sensing, computer vision, artificial intelligence, and interdisciplinary. During the last decade, supervised machine learning has achieved great success and has been successfully applied in the task of ship classification in SAR imagery [1]–[15]. Nevertheless, it still has some limitations for real-world applications due to its strong dependence on massive labeled training data to train a satisfied classifier [16]–[18]. Unfortunately, insufficient labeled training data are an inescapable problem in SAR ship classification tasks. Collecting abundant SAR training data is often expensive, time-consuming, and even unrealistic, which has become one of the most important bottlenecks hindering the further improvement of the supervised learning methods in real SAR ship classification application scenarios.

To handle those problems, several existing methods [19]–[22] introduce the transfer learning (TL) [23]–[25] to deal with the task of ship classification in SAR imagery. TL aims to improve the classification performance or minimize the need for labeled data in a target domain (TD) by leveraging knowledge from one or multiple different but related domain(s) called the source domain (SD). The first attempt came from the work by Lang *et al.* [19], which makes use of the *text messages* of automatic identification system (AIS) as the SD to extract the geometric feature of ships to assist the classification task in the SAR domain, i.e., TD. AIS is designed to automatically provide continuous *text messages*, including static (*name*, *type*, *length*, *width*, and so on) and dynamic (*speed*, *position*, *heading*, and so on) ship information to other ships and coastal authorities. Those messages can be easily accessed and collected from the official website [26] to build a dataset with a large number of labeled instances, thanks to AIS that provides the name, type, and classification feature simultaneously. Lang *et al.* [19] extracted the naive geometric features (NGFs) based on *length* and *width* of ships provided by AIS and proposed an improved multiclass adaptive support vector machine (A-SVM) to achieve knowledge transfer from the AIS domain (SD) to the SAR domain (TD). Their study proved that, by transferring knowledge of the AIS domain, the performance of ship classification can be significantly improved with limited labeled instances (only five in their study) available in the SAR domain. To solve the problem that there are no labeled training data available in SAR domain (TD), Xu *et al.* improved the method of adaptation regularization framework based on transfer learning (ARTL) which was originally proposed by Long *et al.* [27]. Xu *et al.* added a source discriminative information preservation regularization term in the loss function [20] leading to an efficient transfer of interclass discriminative ability from the SD to the TD

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and an efficient distribution alignment between two domains. Furthermore, Xu and Lang [21] proposed the method of geometric transfer metric learning (GTML), which achieved discriminative information preservation, geometric structure preservation, and domain shift adaption simultaneously by embedding pairwise constraints and joint distribution adaptation, together with manifold regularization into a unified optimization process. This method can handle both situations that there are limited or even no labeled instances in the TD. Rostami *et al.* [22] proposed to transfer the knowledge of the high-resolution optical remote sensing (ORS) imagery to the SAR imagery to realize the classification of ships and nonships rather than fine-grained ship classification.

A. Motivation

It is easy to find that the existing studies [19]–[22] merely address the problem of TL from a single SD (AIS domain or ORS domain) to the TD (SAR domain) based on the homogeneous TL (HoTL), which requires both domains to be represented by homogeneous features with the same dimensions [23]–[25]. This status quo inspires us to think about two practical questions.

1) On the Usage of Multiple Source Domains: The existing studies [19]–[22] have fully demonstrated that both AIS and ORS domains can be used as effective SDs to assist ship classification in the SAR domain. Moreover, as the research continues to deepen, more SDs will be discovered. As we know, the AIS domain can provide precise length and width information about a ship because they are read directly from the AIS *text message* itself [26], which is more accurate than those extracted from optical or SAR images by some algorithms. For the ORS domain, in addition to providing basic geometric information, it can also provide rich texture and appearance features (more informative). Therefore, it is reasonable to believe that, by exploring the complementary knowledge of the two or multiple SDs, it is possible to further improve the ship classification performance in the SAR domain.

2) On the Usage of Heterogeneous Features: Since existing studies [19]–[22] are based on HoTL methods, both SD and TD are represented by homogeneous features with the same dimensions. As aforementioned, we can extract precise geometric features from the AIS domain and extract scattering, texture, and even polarization features from the SAR domain, while, for the ORS domain, we can extract more informative image features (especially deep features). It is obvious that each domain has its most appropriate features, and forcing all domains to utilize the exact same features is not conducive to maximizing the unique advantages of the domains. Therefore, it is reasonable to believe that by exploiting more appropriate features for the respective domain (i.e., utilizing heterogeneous features instead of homogeneous features) [28]; it is possible to further improve the ship classification performance in the SAR domain.

In view of the above, for the purpose of improving the ship classification performance in the SAR domain even further, this study takes a step forward and attempts to develop a multisource heterogeneous TL (MS-HeTL) method that conducts TL from multiple SDs and allows different domains to be represented as heterogeneous features with different dimensions.

B. Challenge

To develop the MS-HeTL method, there are two immediate challenges that must be properly addressed.

1) Feature Heterogeneity: The HoTL that is widely used by the existing methods [19]–[22] utilizes homogeneous features with the same dimensions to transfer knowledge between domains. A typical advantage of HoTL is that both the SD(s) and the TD use the same feature that can preserve the relevance between two domains in a common feature space, thereby avoiding the negative transfer [29]. When utilizing heterogeneous features with different dimensions, the heterogeneous TL (HeTL) [28] must consider both distribution adaption to minimize the domain shift and feature space adaption to eliminate the feature heterogeneity, which makes developing an HeTL algorithm more complicated than developing an HoTL algorithm.

2) Domain Discrepancy: This study aims to transfer the knowledge of both AIS and ORS domains to assist ship classification tasks in the SAR domain. Another immediate challenge comes from the different nature of the three domains: AIS is *text messages*, which is completely different from ORS and SAR imagery. Although both ORS and SAR are imagery, they have completely different imaging mechanisms (passive versus active), spatial resolution (submeter versus meter), and data structures (panchromatic versus polarization) [30], [31]. The wide gap between three domains leads to heterogeneous feature spaces that exist not only between the SDs and TD, but also between SDs each other. In our case study, the feature extracted from AIS is NGFs [5] and from ORS is deep feature (DeCAF_6 [32]), whereas from SAR is another deep feature (ResNet_{18} [33]). In the literature, there are very few approaches to deal with the problem of MS-HeTL. One representative method is domain adaptation with manifold alignment (DAMA) proposed by Wang and Mahadevan [34]. However, DAMA is not suitable for dealing with multiple heterogeneous SDs because it treats each SD equally and does not distinguish the difference in importance between domains. Although several deep learning methods have been developed for multisource TL, they can only deal with HoTL tasks but are inapplicable to handle HeTL tasks because they make a strong assumption that all the domains have the same feature representation [35]–[38].

C. Contribution

The major contributions of this study are threefold as follows.

1) Novel Method: In this study, we propose multisource heterogeneous feature augmentation (MS-HFA) that is an extension of heterogeneous feature augmentation (HFA) [39] for the single SD case to handle the problem of MS-HeTL. MS-HFA can effectively solve the challenges brought by

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feature heterogeneity and fully exploit the complementarity cross domains. In addition, we also specially design a support vector machine (SVM) classification framework to incorporate the augmented feature representations in the common space to conduct the knowledge transfer cross domains.

2) New Dataset: For research purposes, this study collects a new ORS dataset. The dataset contains eight ship classes with a total of 8678 pictures. All pictures are collected using Google Earth with submeter resolution, and corresponding class information is matched with the official website [26]. Unlike some existing ORS ship datasets [40], [41], whose single picture may contain multiple ships or even incomplete ships, the imagery slices in our dataset are segmented more carefully. Each slice contains only one ship, so the users can focus only on the evaluation of the classification method, without the need to perform additional processing on the pictures. We sincerely hope that it becomes new data support for the community.

3) Extensive Research: To demonstrate the effectiveness of the proposed method, we conduct extensive experiments to compare the proposed method to: 1) several baseline methods including traditional classifiers and classical convolutional neutral network classifiers; 2) HeTL methods using single SD; 3) multisource HoTL (MS-HoTL); and 4) the state-of-the-art MS-HeTL.

The new dataset that we contribute and all source codes related to the proposed method have been released at https://github.com/BUCT-RS-ML/MS-HeTL-via-MS-HFA.

The remainder of this article is organized as follows. Section II describes the proposed method in detail, including problem definition, the proposed MS-HFA, and how to incorporate it into the SVM training framework. Section III introduces the experimental protocol, including datasets, features, comparison methods, implementation details, and evaluation criteria. In Section IV, we report the results of four comparison experiments with the analysis based on the confusion matrix; we also report the experimental results that the method is addressing semisupervised situations. Finally, we conclude and outlook our study in Section V. The detailed methods related to how to incorporate MS-HFA with the SVM training framework and how to extend the proposed method to semisupervised situations are appended in Appendixes A and B, respectively.

II. Methodology

On the assumption that there are only one SD and one TD, Li *et al.* [39] and Duan *et al.* [42] proposed HFA to deal with the heterogeneous domain adaptation (HDA) problem. In this study, we extend HFA to MS-HFA to address the MS-HeTL problem for the purpose of improving ship classification performance in the SAR domain by utilizing complementary knowledge from multiple heterogeneous SDs.

A. Problem Formulation and Notations

Suppose that we have m SDs, $\mathcal{D}_{s1}, \ldots, \mathcal{D}_{si}, \ldots, \mathcal{D}_{sm}$ with abundant labeled instances, and we define the ith SD $\mathcal{D}_{si}$ with n_{si} instances as $\{(\mathbf{x}_{si}^j, y_{si}^j)\}_{j=1}^{n_{si}}$, where $\mathbf{x}_{si}^j \in \mathbb{R}^{d_{si}}$ denotes the feature column vector of the instance in $\mathcal{D}_{si}$ and y_{si}^j is the corresponding label. Simultaneously, we have one TD $\mathcal{D}_t$ with n_t ($n_t \ll n_{si}$) labeled instances as $\{(\mathbf{x}_t^j, y_t^j)\}_{j=1}^{n_t}$, where $\mathbf{x}_t^j \in \mathbb{R}^{d_t}$ denotes the feature column vector of the TD labeled instance, respectively, and y_t^j is the corresponding label of $\mathbf{x}_t^j$. Note that both SDs and TD share the same label space.

The TL aims to learning the knowledge from one or multiple SD(s) to improve the task performance on the TD. If $m \geq 2$ and condition $d_{s1} = \cdots = d_{sm} = d_t$ is satisfied, it is regarded as an MS-HoTL problem [43]. However, if $m \geq 2$ and the above condition is not satisfied, that is to say, the feature dimensions of different domains are not exactly the same, it is defined as an MS-HeTL problem that is the topic this study focuses on, while, when $m = 1$, the above definition can be simplified to single-source TL situation. In this situation, the single SD $\mathcal{D}_s$ with n_s labeled instances can be easily represented as $\{(\mathbf{x}_s^j, y_s^j)\}_{j=1}^{n_s}$, where $\mathbf{x}_s^j \in \mathbb{R}^{d_s}$ and y_s^j are the corresponding labels. Similarly, single-source TL can be divided into HoTL when $d_s = d_t$ (SS-HoTL) and HeTL when $d_s \neq d_t$ (SS-HeTL).

B. Heterogeneous Feature Augmentation

To handle that the data from the SD and the TD are represented by heterogeneous features ($\mathbf{x}_s \neq \mathbf{x}_t$) with different dimensions ($d_s \neq d_t$), Li *et al.* [39] and Duan *et al.* [42] proposed HFA. They first project $\mathbf{x}_s$ and $\mathbf{x}_t$ into a d_c-dimensional common subspace by using two different feature projection matrices $\mathbf{P}_s \in \mathbb{R}^{d_c \times d_s}$ and $\mathbf{P}_t \in \mathbb{R}^{d_c \times d_t}$, respectively. In the common subspace, the similarity of heterogeneous features from two domains can be measured and compared. Then, two feature mapping functions φ_s and φ_t, as illustrated in (1), are proposed to augment the transformed features in the common subspace. By this way, both the transformed features ($\mathbf{P}_s\mathbf{x}_s$ and $\mathbf{P}_t\mathbf{x}_t$) and the original features ($\mathbf{x}_s$ and $\mathbf{x}_t$), together with zeroes column vectors ($\mathbf{0}_{d_t}$ and $\mathbf{0}_{d_s}$), are incorporated into an augmented common feature space for original information preservation, making the heterogeneous features from two domains linked together

$$\varphi_s(\mathbf{x}_s) = \begin{bmatrix} \mathbf{P}_s\mathbf{x}_s \\ \mathbf{x}_s \\ \mathbf{0}_{d_t} \end{bmatrix} \quad \text{and} \quad \varphi_t(\mathbf{x}_t) = \begin{bmatrix} \mathbf{P}_t\mathbf{x}_t \\ \mathbf{0}_{d_s} \\ \mathbf{x}_t \end{bmatrix}. \tag{1}$$

With the augmented features, Li *et al.* [39] and Duan *et al.* [42] proposed to find the two projection matrices $\mathbf{P}_s$ and $\mathbf{P}_t$ for the source and target data by using the standard SVM training framework in both linear and nonlinear cases.

C. Multisource Heterogeneous Feature Augmentation

We extend HFA to MS-HFA to address the multisource scenario. As shown in Fig. 1, we first extract feature representations $\mathbf{x}_{s1}, \mathbf{x}_{s2}, \ldots, \mathbf{x}_{sm}, \mathbf{x}_t$ from respective domains including both SDs and TD. Next, we project all these feature representations into a d_c-dimensional common subspace. The corresponding projection matrices for ith SD $\mathcal{D}_{si}$ are denoted as $\mathbf{P}_{si} \in \mathbb{R}^{d_c \times d_{si}}$ and for TD $\mathcal{D}_t$ as $\mathbf{P}_t \in \mathbb{R}^{d_c \times d_t}$. Then, we build an augmented common feature space in the multisource scenario by using m feature augmentation projections $\varphi_{s1}, \ldots, \varphi_{sm}$ for

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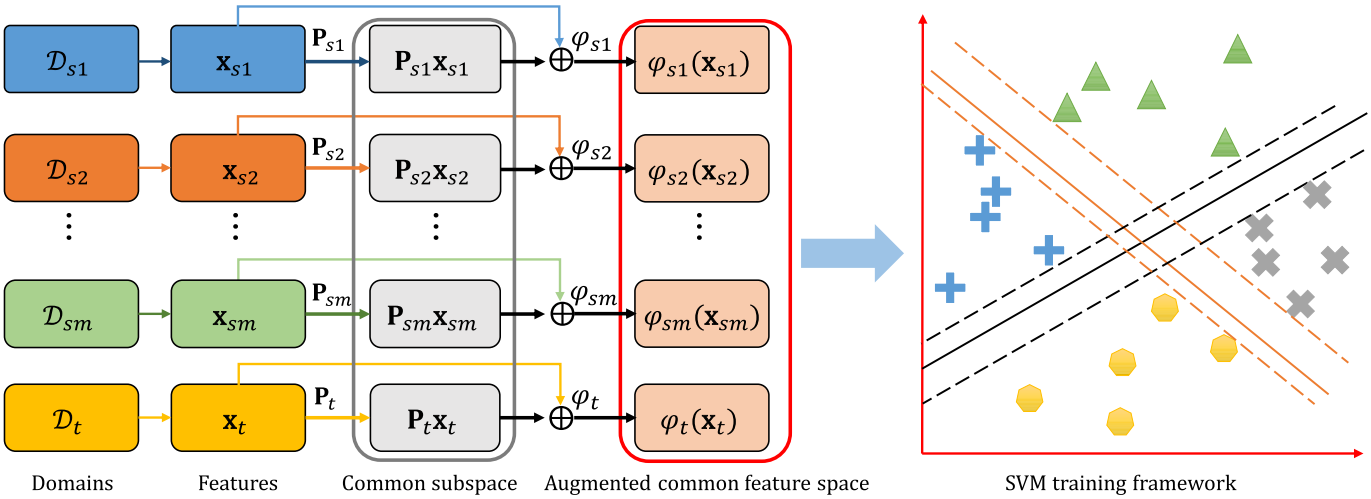


Fig. 1. Flowchart of MS-HFA incorporated with the SVM training framework. The proposed method is composed of two steps. In the first step, heterogeneous features $\mathbf{x}_{s1}, \mathbf{x}_{s2}, \ldots, \mathbf{x}_{sm}, \mathbf{x}_t$ extracted from respective domains (including both SDs and TD) are projected into a common subspace by using corresponding projection matrices $\mathbf{P}_{s1}, \mathbf{P}_{s2}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t$, respectively, and then, the transformed features in the common subspace and the original features together with zeroes vectors are further transformed into an augmented common feature space by using the feature augmentation transformation $\varphi_{s1}, \varphi_{s2}, \ldots, \varphi_{sm}, \varphi_t$ [refer to (2) and (3)]. By this way, the similarities between data from the same domain are enhanced. In the second step, a specific-designed multiple-class SVM training framework is incorporated with the augmented feature representations $\varphi_{s1}(\mathbf{x}_{s1}), \varphi_{s2}(\mathbf{x}_{s2}), \ldots, \varphi_{sm}(\mathbf{x}_{sm}), \varphi_t(\mathbf{x}_t)$ to learn the projection matrices $\mathbf{P}_{s1}, \mathbf{P}_{s2}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t$. By this way, we can learn individual model for each ship class.

SDs and one φ_t for TD, as defined in (2) and (3), respectively. The strategy of MS-HFA has two advantages. The first is that this transformation can preserve the original information for respective domain and can enhance the similarities between data from the same domain. The second, more important, is that this transformation eliminates the feature heterogeneity between domains by projecting heterogeneous features of all domains into an augmented common feature space with the same dimensions

$$\varphi_{s1}(\mathbf{x}_{s1}) = \begin{bmatrix} \mathbf{P}_{s1}\mathbf{x}_{s1} \\ \mathbf{x}_{s1} \\ \cdots \\ \mathbf{0}_{d_{sm}} \\ \mathbf{0}_{d_t} \end{bmatrix}, \ldots, \varphi_{sm}(\mathbf{x}_{sm}) = \begin{bmatrix} \mathbf{P}_{sm}\mathbf{x}_{sm} \\ \mathbf{0}_{d_{s1}} \\ \cdots \\ \mathbf{x}_{sm} \\ \mathbf{0}_{d_t} \end{bmatrix} \tag{2}$$

and

$$\varphi_t(\mathbf{x}_t) = \begin{bmatrix} \mathbf{P}_t\mathbf{x}_t \\ \mathbf{0}_{d_{s1}} \\ \cdots \\ \mathbf{0}_{d_{sm}} \\ \mathbf{x}_t \end{bmatrix}. \tag{3}$$

D. Incorporating Into SVM Training Framework

Li *et al.* [39] and Duan *et al.* [42] proposed to learn the projection matrices $\mathbf{P}_s$ and $\mathbf{P}_t$ by using the standard SVM with the hinge loss function. We employ a similar strategy that learns the feature projection matrices $\{\mathbf{P}_{si}\}_{i=1}^m$ and $\mathbf{P}_t$ by incorporating MS-HFA into SVM training framework. It should be noted that the following solution that we proposed is not a simple duplication to [39] and [42]. From HFA to MS-HFA, we not only have to consider the surge in solution complexity due to the increase in the number of SDs but also particularly consider the problem of domain weight assignment in the process of MS-HeTL (this is a case that HFA has not studied). The solution process is introduced as follows, and the detailed derivation is described in Appendix A.

For the augmented common feature space, we define feature weight vector $\mathbf{w} = [\mathbf{w}_c^T, \mathbf{w}_{s1}^T, \ldots, \mathbf{w}_{sm}^T, \mathbf{w}_t^T]^T$, where $\mathbf{w}_c \in \mathbb{R}^{d_c}$, $\{\mathbf{w}_{si} \in \mathbb{R}^{d_{si}}\}_{i=1}^m$, and $\mathbf{w}_t \in \mathbb{R}^{d_t}$ are the feature weight vectors designed for the common subspace, $\mathcal{D}_{si}$, and $\mathcal{D}_t$, respectively. The superscript $(\cdot)^T$ denote the transpose operation. We learn the projection matrices $\{\mathbf{P}_{si}\}_{i=1}^m, \mathbf{P}_t$ and the feature weight vector $\mathbf{w}$ by minimizing the structural risk and the empirical risk of SVM with hinge loss. The optimization formulation of MS-HFA in the linear case is defined as follows:

$$\min_{\mathbf{P}_{si},\mathbf{P}_t} \min_{\mathbf{w},b,\xi_{si}^j,\xi_t^j} \frac{1}{2}\mathbf{w}^T\mathbf{w} + C\left(\sum_{i=1}^{m}\theta_{si}\sum_{j=1}^{n_{si}}\xi_{si}^j + \theta_t\sum_{j=1}^{n_t}\xi_t^j\right) \tag{4}$$

$$\text{s.t.} \left\{y_{si}^j\left(\mathbf{w}^T\varphi_{si}\left(\mathbf{x}_{si}^j\right) + b\right) \geq 1 - \xi_{si}^j, \xi_{si}^j \geq 0\right\}_{i=1}^m \tag{5}$$

$$y_t^j\left(\mathbf{w}^T\varphi_t\left(\mathbf{x}_t^j\right) + b\right) \geq 1 - \xi_t^j, \quad \xi_t^j \geq 0 \tag{6}$$

$$\left\{\|\mathbf{P}_{si}\|_F^2 \leq \lambda_{si}\right\}_{i=1}^m, \quad \|\mathbf{P}_t\|_F^2 \leq \lambda_t \tag{7}$$

where the term before the first "+" in (4) is the structural risk to describe the model complexity and the reminder term is empirical losses on the training instances from all domains. $C > 0$ is used to tradeoff those two terms. In particular, it should be pointed out that, considering the different contributions of various domains in the augmented common feature space, we assign domain weight parameters $\theta_{si} > 0$ and $\theta_t > 0$ to quantify the contribution of $\mathcal{D}_{si}$ and $\mathcal{D}_t$, and introduce slack variables ξ_{si}^j and ξ_t^j for the jth instance in $\mathcal{D}_{si}$ and $\mathcal{D}_t$, respectively. In (5) and (6), b denotes the bias term of separating hyperplane. In (7), $\|\cdot\|_F$ means the Frobenius norm of a matrix, and $\lambda_{si}, \lambda_t > 0$ are predefined parameters

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to constrain the complexity of the corresponding projection matrix.

To solve the optimization problem in (4), we first introduce a series of dual variables $\alpha_{si}^1, \ldots, \alpha_{si}^{n_{si}}$ for instances in $\mathcal{D}_{si}$ and $\alpha_t^1, \ldots, \alpha_t^{n_t}$ for instances in $\mathcal{D}_t$, and then, we use the Lagrangian multiplier method to derive its dual form. To simplify the nontrivial optimization problem, we introduce an intermediate transformation matrix $\mathbf{H} = [\mathbf{P}_{s1}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t]^T [\mathbf{P}_{s1}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t] \in \mathbb{R}^{d \times d}$ to combine the feature projection matrices. By this way, we only need to solve for an individual $\mathbf{H}$ rather than $\{\mathbf{P}_{si}\}_{i=1}^m$ and $\mathbf{P}_t$. Then, we generalize the dual problem to nonlinear situation for its kernelization and further define a nonlinear transformation matrix $\widehat{\mathbf{N}} \in \mathbb{R}^{n \times n}$ to equivalently replace the linear $\mathbf{H}$. $\widehat{\mathbf{N}}$ satisfies $\mathbf{H} = \Psi \mathbf{K}^{-(1/2)} \widehat{\mathbf{N}} \mathbf{K}^{-(1/2)} \Psi^T$ [44], and the size of $\widehat{\mathbf{N}}$ is independent to feature dimension. Finally, we reformulate the nonlinear dual problem to a convex multiple kernel learning (MKL) problem [45] by decomposing $\widehat{\mathbf{N}}$ into a linear combination of a series of rank-one normalized positive semidefinite (PSD) matrices, i.e., $\widehat{\mathbf{N}} = \sum_{r=1}^{\infty} \eta_r \mathbf{Z}_r$, which can use the existing MKL solvers [46] to obtain the global solution.

After obtaining the optimal solution of vector $\boldsymbol{\alpha} = [\alpha_{s1}^1, \ldots, \alpha_{s1}^{n_{s1}}, \ldots, \alpha_{sm}^1, \ldots, \alpha_{sm}^{n_{sm}}, \alpha_t^1, \ldots, \alpha_t^{n_t}]^T$ and transformation $\widehat{\mathbf{N}}$ based on the MKL solver, we can obtain the target decision function for any unlabeled instance $\mathbf{x}$ in the TD as

$$f(\mathbf{x}) = (\boldsymbol{\alpha} \circ \mathbf{y})^T \mathbf{K}^{\frac{1}{2}} \left(\widehat{\mathbf{N}} + \mathbf{E}_n\right) \begin{bmatrix} \mathbf{O}_{(n-n_t) \times n_t} \\ \mathbf{K}_t^{-\frac{1}{2}} \end{bmatrix} \mathbf{k}_t + b \tag{8}$$

where $\boldsymbol{\alpha} \circ \mathbf{y}$ means the Hadamard product between two vectors. $\mathbf{E}_n$ means the $n \times n$ identity matrix, and n means the total number of all training instances. $\mathbf{k}_t = [k(\mathbf{x}_t^1, \mathbf{x}), \ldots, k(\mathbf{x}_t^{n_t}, \mathbf{x})]^T$, and $k(\cdot, \cdot)$ is a predefined kernel function for two instances from the TD.

We use libSVM as the basic SVM classifier [47] and follow the strategy of "one versus one" to construct a multiclass classifier [48]. The test instances will be submitted to all subclassifiers simultaneously, and then, the majority voting strategy is applied to determine the final prediction.

The above process is only a general description of the proposed solution, and interested readers can refer to Appendix A for specific derivation details.

III. Experimental Protocol

A. Datasets

In this study, we conducted a ship classification task on two public SAR imagery datasets **HR-SAR** [2] and **FUSAR** [49], respectively, which are regarded as the TDs. At the same time, we took **AIS** [19] and **ORS** datasets as the SDs to assist the ship classification task in the SAR domain. Some sample data in these four datasets are shown in Fig. 2. The details of those datasets are described as follows.

The first SD **AIS** dataset was collected by Lang *et al.* [19]. This dataset contains four ship classes, i.e., cargo, bulk carrier, container, and oil tanker, and each class has 600 labeled samples. Each sample in the dataset is essentially a piece of *text message* that corresponds to a specific ship [as shown in Fig. 2(a)], specifying the maritime mobile service identify (MMSI), class, length and width that have been used to extract the NGFs by existing work [19]–[21] to bridge the AIS and SAR domains.

MMSI	Class	Length (m)	Width (m)
412434970	Cargo	88	13
413551110	Cargo	95	15
413733000	Bulk carrier	225	32
413095000	Bulk carrier	289	45
636091716	Container	264	32
636014997	Container	306	40
412378820	Oil tanker	151	21
413180000	Oil tanker	333	60

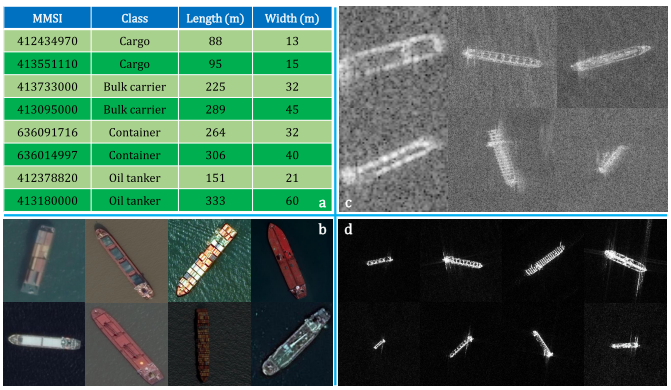


Fig. 2. Sample data in four datasets: (a) AIS, (b) ORS, (c) HR-SAR, and (d) FUSAR.

Another SD **ORS** dataset was specially built by authors for study purposes. The dataset has eight ship classes, including cargo, container, oil tanker, bulk carrier, car carrier, chemical tanker, dredge, and tug. All ship class information was matched with the official website [26]. All the optical imageries were collected from Google Earth with submeter resolution. Unlike some existing ORS ship datasets [40], [41], whose single slice may contain multiple ships or even incomplete ships, the image slices in our dataset are segmented more carefully, and each slice is interpreted collaboratively by multiple experts. Each picture contains only one ship, as shown in Fig. 2(b), so the users can focus only on the evaluation of the classification method, without the need to perform additional processing on the pictures. In this study, we chose a subset that contains four common ship classes as the **AIS** dataset and also made use of 600 labeled samples per class.

The **HR-SAR** is a high-resolution TerraSAR-X SAR ship dataset released by Xing *et al.* [2]. This dataset contains three ship classes, i.e., cargo, container, and oil tanker, and has 50 samples per class.

The **FUSAR** dataset was collected by Hou *et al.* [49] from high-resolution Gaofen-3 imagery. In this study, we chose a subset that contains four common ship classes, i.e., cargo, bulk carrier, container, and oil tanker, from the original dataset to match the ship classes to SD datasets. Moreover, in order to avoid the effect caused by class imbalance, we used 50 samples for each class.

In our experiments, we performed three- and four-class ship classification tasks on the **HR-SAR** and **FUSAR** datasets, respectively, while transferring the knowledge of the two SDs, **AIS** and **ORS**, to assist the classification in the SAR domain. As aforementioned, this study assumed that there were only limited labeled samples in the TD. Thus, we made use of three labeled samples per class to train the target classifier and test the performance of the rest samples (47 per class). As for the SD, we made use of 20 labeled samples per class to conduct MS-HeTL. For clarity, Table I summarizes the data usage statistics in our experiments.

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TABLE I
DATA AND FEATURE USAGE STATISTICS

Dataset	Type of data	Number of classes	Number of samples per class	Number of labeled samples used per class	Feature	Dimension
AIS	Text message	4	600	20	NGFs	11
ORS	Optical imagery	4	600	20	DeCAF_6	4096
HR-SAR	SAR imagery	3	50	3	ResNet_{18}	512
FUSAR	SAR imagery	4	50	3	ResNet_{18}	512

B. Features

For the AIS domain, we followed the existing works [19]–[21] and extracted 11-D NGFs [5] as the feature representations. For the ORS domain, we took the 4096-D DeCAF_6 [32] as the feature representations, which has been widely used in recent image classification tasks. For HR-SAR and FUSAR domains, we took ResNet18 [33] network as the feature extractor to extract 512-D ResNet_{18} as the feature representations. The detailed features usage in our experiments is listed in Table I.

It should be noted that the goal of this study is not to achieve the best ship classification performance by exploring the most appropriate feature representations for respective domains. Our aim is to demonstrate that the proposed method can improve the effectiveness of knowledge transfer compared to existing methods under the premise of using common data and features. We believe that, with the progress of feature representation studies, the TD classification performance is destined to be further improved.

C. Comparison Methods

In order to evaluate the proposed method comprehensively, we compared it to state-of-the-art methods. These methods and their brief descriptions are listed in Table II. In general, these methods can be divided into four categories: baseline methods, SS-HeTL methods (SS-HeTL), MS-HoTL, and MS-HeTL.

By comparing with baseline methods, we aim to demonstrate that the proposed method does not suffer from negative transfer and can achieve better performance than directly training classifiers (whether traditional or deep learning-based) on the TD. SVM_t [47] was trained with three labeled samples in the TD. Three deep learning-based methods, AlexNet [50], VGG16 [51], and ResNet18 [33], were pretrained on the ImageNet dataset [60] first and then fine-tuned [61] in the TD. We did not use more complex neural networks because both theory and our preliminary experiments show that more complex networks (deeper and/or wider) suffer from severe overfitting when there are only three training samples per class available in the SAR domain [62], [63].

We compared the proposed method with SS-HeTL methods to demonstrate that, compared to using a single SD, using more SDs may help to improve the TD classification performance. To this end, we chose a series of typical HeTL methods working for a single SD. These methods include max-margin domain transforms (MMDT) [52], HFA [39], cross-domain landmark selection (CDLS) [53], iterative reweighting heterogeneous transfer learning (IRHTL) [54], and cross-domain structure preserving projection (CDSPP) [55]. The details are described in Table II.

We also compared the proposed method with MS-HoTL methods to demonstrate that, compared with the use of homogeneous features in all domains, the use of heterogeneous features in different domains is more conducive to improving the classification performance for the task of ship classification in SAR imagery. The A-SVM [19] classifier was used to conduct MS-HoTL.

Finally, we compared two state-of-the-art MS-HeTL methods, DAMA [34] and conditional weighting adversarial network (CWAN) [59], to evaluate the effectiveness of the proposed method. Noting that, although CWAN is essential a semisupervised method, we still took it as a comparison method due to the scarcity of MS-HeTL methods.

D. Implementation Details

For the comparison methods listed in Table II, we report the best results that we can achieve by parameters tuning within their recommended range in their original literature. Except for CWAN that uses softmax as the classifier, other methods use SVM with radial basis function (RBF) kernel as the classifier and use the grid search technique to choose the best parameters, i.e., the tradeoff parameter C and the bandwidth parameter γ in a large search space as $\{0.01, 0.1, 1, 10, 100\}$. For the proposed method, we empirically set $\lambda = 1000$ and search in the range of $\{0.1, \ldots, 0.9, 1.0\}$ for the optimal weight of SDs (θ_{s1} and θ_{s2}) while fixing the weight of TD as $\theta_t = 1$.

All the experiments are run on an Intel[1] Core[2] i7-6800K 3.40-GHz CPU with Windows10 64-bit operating system and 64-GB RAM.

E. Evaluation Criteria

To gain a statistically meaningful understanding, the experiment is repeat ten times to eliminate the possible bias caused by random sampling of data in each individual experiment. The average performance, i.e., mean accuracy and standard deviation (acc. $\pm$ std.) over ten runs, is reported to evaluate the performance of various methods. In each individual experiment, we define the classification accuracy as

$$\text{accuracy} = \frac{\left|\left\{\mathbf{x} : \mathbf{x} \in \mathfrak{D} \bigwedge f(\mathbf{x}) = y(\mathbf{x})\right\}\right|}{\left|\{\mathbf{x} : \mathbf{x} \in \mathfrak{D}\}\right|} \times 100\%$$

where $\mathfrak{D}$ denotes the test dataset. $\mathbf{x}$ is the instance in $\mathfrak{D}$, and $f(\mathbf{x})$ and $y(\mathbf{x})$ denote the predicted label and the ground-truth label of $\mathbf{x}$, respectively.

[1]Registered trademark.
[2]Trademarked.

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TABLE II
COMPARISON METHODS

Category	Methods	Brief description
Baselines	SVM_t [47]	SVM_t is trained only with three labeled samples in the target domain without any auxiliary from the source domains. [47]
	AlexNet [50]	AlexNet is first pre-trained on the ImageNet dataset and is then fine-tuned by only using three labeled samples in TD. [50]
	VGG16 [51]	VGG16 is first pre-trained on the ImageNet dataset and is then fine-tuned by only using the labeled samples in TD. [51]
	ResNet18 [33]	ResNet18 is first pre-trained on the ImageNet dataset and is then fine-tuned by only using the labeled samples in TD. [33]
SS-HeTL	MMDT [52]	MMDT is a supervised method which learn a asymmetric transformation from TD to aligned with SD. The main idea behind this method is to simultaneously learn both the transformation and the classifier parameters themselves, and use the same classification loss to jointly optimize both of them. [52]
	HFA [39]	HFA makes two transformations for each domain via feature augmentation technique to map the original features into an augmented feature space, then learns the transformations by incorporating with the SVM training framework. The details is described in Section II-B.
	CDLS [53]	CDLS firstly constructs a projection matrix to map SD data into the derived subspace of TD so that cross-domain data can be matched, and then perform distribution alignment and landmark selection to iterative update the projection. Noting that in the experiments, we turn CDLS into its supervised version for a fair comparison. [53]
	IRHTL [54]	IRHTL firstly transforms the SD and TD data into a common space by using two domain-specific projections, then base on SVM training framework to iteratively learn the optimal transforms and reweight the SD instances. [54]
	CDSPP [55]	CDSPP extends classic locality preserving projection (LPP) [56] to construct two domain-specific projections to map features into a common subspace so that the class consistency is preserved and feature distributions are aligned. [55]
MS-HoTL	A-SVM [19]	A-SVM [57] can adapt one or multiple SDs SVM classifiers to a SVM classifier by learning "delta function" between the decision functions of the auxiliary and TD classifiers by using an objective function extended from single source A-SVMs [58]. Literature [19] further improve the method by introducing a regulation parameter Γ to balance the margin maximization and knowledge transfer.
MS-HeTL	DAMA [34]	DAMA is designed for MS-HeTL problem. It aims to learn mapping functions for each input domain (contains SDs and TD) to map all domains data into a latent common space where the manifolds of data from all domains are aligned and the topology of each domain is preserved. [34]
	CWAN [59]	CWAN is also designed for MS-HeTL problem, it learns feature transformations for each input domain to map the data of all domains into a common space by using the network, and then based on multi-source conditional weighting schemes to perform adversarial distribution alignment between each SD and TD. [59]

TABLE III
PERFORMANCE OF DIFFERENT METHODS ON THE HR-SAR AND FUSAR DATASETS (ACC. ± STD. %)

Category	Methods	SDs	TDs	
			HR-SAR	FUSAR
Baseline	SVM_t	None	80.71 ± 3.39	65.90 ± 6.95
	AlexNet		76.83 ± 2.49	65.81 ± 4.02
	VGG16		77.30 ± 0.71	68.26 ± 4.02
	ResNet18		79.19 ± 0.41	71.81 ± 2.32
SS-HeTL	MMDT	AIS	57.66 ± 8.54	49.47 ± 3.81
	HFA		83.19 ± 3.31	72.71 ± 3.96
	CDLS		83.12 ± 2.56	67.94 ± 6.82
	IRHTL		83.97 ± 2.72	72.39 ± 4.02
	CDSPP		83.83 ± 3.20	71.01 ± 4.74
	MMDT	ORS	83.55 ± 3.44	72.45 ± 4.59
	HFA		83.62 ± 2.51	71.06 ± 4.15
	CDLS		82.06 ± 2.97	70.07 ± 6.46
	IRHTL		83.97 ± 2.59	71.17 ± 4.47
	CDSPP		83.69 ± 2.55	72.18 ± 4.07
MS-HoTL	A-SVM	AIS & ORS	81.84 ± 5.04	58.14 ± 4.53
MS-HeTL	DAMA		82.62 ± 3.68	68.35 ± 4.06
	CWAN		82.70 ± 2.72	70.48 ± 6.12
	Proposed		**85.60 ± 2.63**	**74.20 ± 3.92**

IV. RESULTS AND DISCUSSION

A. Experimental Results

The experimental results on two SAR datasets **HR-SAR** and **FUSAR** are listed in Table III.

1) Comparing With Baselines: We find that two types of baseline methods, traditional SVM_t and three deep learning-based methods, exhibit large differences in the two SAR datasets. On the relatively simple HR-SAR dataset (three classes), SVM_t outperforms deep learning-based methods. In contrast, deep learning-based methods outperform SVM_t on the FUSAR dataset. It shows that deep learning-based methods are more suitable for handling complex classification tasks because they can extract more discriminative feature representations than traditional methods. We also find that the proposed method outperforms all baseline methods on both datasets. Specifically, the proposed method outperforms SVM_t (80.71%), which is the best baseline result on the HR-SAR dataset by 4.89%, and outperforms ResNet18 (71.81%) on the FUSAR dataset by 2.39%. This result demonstrates that, when only a very small number of labeled samples are available in the SAR domain, the proposed method is better than the widely used methods, that is, fine-tuning pretrained CNN classifiers. Therefore, it is valuable to utilize TL to help improve the performance of ship classification in SAR imagery.

2) Comparing With SS-HeTL Methods: Observing the performance of the SS-HeTL methods on two SAR datasets and comparing with the proposed method, we find that, on the HR-SAR dataset, compared with baseline methods, the performance of SS-HeTL methods is better, except for

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max-margin domain transforms (MMDT) with AIS as the SD (57.66%). On the relatively difficult FUSAR dataset, except that the performance of MMDT and CDLS with AIS as the SD (49.47% and 67.94%) is poor than that of VGG16 (68.26%) and ResNet18 (71.81%), the performance of other SS-HeTL methods achieves comparable or even better performance than that of baseline methods. The above comparison shows that the performance of SS-HeTL is directly affected by the characteristics of the SD. In some cases, it is even better to train the classifier directly on the TD. However, the above situation does not appear in the proposed MS-HeTL method whose performance outperforms all baseline methods and SS-HeTL methods on both SAR datasets. On the HR-SAR dataset, it is 1.63% higher than the best result obtained by IRHTL with AIS or ORS as SD (83.97%). On the FUSAR dataset, it is 1.49% higher than the best result obtained by HFA with AIS as SD (72.71%). This shows that the proposed method is, indeed, able to extract richer and complementary knowledge from the two SDs and perform the effective transfer, thereby steadily improving the classification performance.

3) Comparing With MS-HoTL Methods: It is easy to find that the proposed method completely crushes A-SVM, which is an MS-HoTL method. The result is reasonable because MS-HoTL restricts all domains to use the exact same features (NGFs in our experiments[3]). For the AIS domain, the NGFs may be suitable, but, for the ORS and SAR domains, its discriminative ability is obviously not enough. This point can also be evidenced by the fact that the performance of A-SVM is even inferior to that of most SS-HeTL methods. This experiment proves that, for the ship classification task in SAR imagery, it is clearly more appropriate to use heterogeneous features for different domains, and the proposed method can handle exactly this problem.

4) Comparing With MS-HeTL Methods: In the last experiment, we compared two state-of-the-art MS-HeTL methods, DAMA and CWAN, to evaluate the effectiveness of the proposed method. The results show that DAMA and CWAN are significantly better than the MS-HoTL method (especially on FUSAR data) but still lag behind the proposed method by about 3% on the HR-SAR dataset and 4% (CWAN)–6% (DAMA) on the FUSAR dataset. The results also show that the performance of DAMA and CWAN is even slightly lower than the SS-HeTL methods in most cases, which indicates that MS-HeTL is by no means a trivial task. When dealing with multiple domains with large differences, the method must be able to carefully address both feature heterogeneity and domain discrepancy. The theoretical analysis shows that DAMA treats the data from different domains equally so that it cannot differentiate the importance of knowledge from multiple heterogeneous domains. CWAN takes the conditional weighting schemes into account to distinguish the importance of individual domains, but it may ignore the preservation information of original features during the multilayer feature transformation process. The proposed method considers both domain weight assignment and original features information preservation and, thereby, obtains the highest performance.

[3]Since the AIS domain can only provide NGFs feature, we used it for all domains.

B. Confusion Matrix Analysis

To deeply analyze the reasons behind the experimental results listed in Table III, we employ a confusion matrix analysis method to explore why the proposed method can obtain better performance than all the compared methods. We take the experiments on the FUSAR dataset as a showcase and arrange the confusion matrices of some representative methods in Fig. 3.

Observing the classification performance of a specific ship class, it shows that, in general, the "cargo" class can be correctly classified by most methods (except CWAN, all other methods obtain more than 80% accuracy). Although a bit harder than the "cargo" class, most methods achieve or exceed 75% accuracy on the "bulk carriers" class (A-SVM is an exception). The "container" class is the most difficult to be distinguished from other ship classes, and most methods can not achieve an accuracy of more than 55%. These methods are most divergent on the "oil tanker" class with the best (the proposed method 77.87%) and worst (A-SVM 42.55%) results differing by more than 35% although the proposed method does not perform as well as some methods on two relatively tractable classes. For example, for "cargo" class, it does not outperform ResNet18, HFA (AIS $\rightarrow$ SAR), and MMDT (ORS $\rightarrow$ SAR). For the "bulk carrier" class, it is also inferior to ResNet18 and MMDT (ORS $\rightarrow$ SAR). However, it far outperforms all opponents in two intractable classes. That is, for the "container" class, it exceeds the second place (CWAN) by 3.19% (59.79% versus 56.60%), and for the "oil tanker" class, it not only ranks first but even leads A-SVM by more than 35%. This shows that the proposed method has the most balanced performance cross ship classes, which can be found in Table IV. In Table IV, the mean and standard deviation of accuracy four ship classes (i.e., the diagonal elements of each confusion matrix in Fig. 3) are calculated. It is obvious that the mean here is in one-to-one correspondence with Table III, and the standard deviation can reflect the class equilibrium of comparison methods. The smaller the value, the higher the balance ability, and vice versa. Table IV shows that the deviation of the proposed method (10.19%) is smaller than all methods except CWAN (9.50%, whereas its mean of 70.48% is much lower than that of the proposed method 74.20%), which demonstrates its balancing ability. The above analysis indicates that, through the proposed MS-HFA operation, in the augmented common feature space, the intraclass samples become more compact, while the interclass samples are further pulled apart, and the SVM finds the better balanced hyperplanes to separate the different ship classes, thereby improving the overall classification accuracy.

C. Extension to Semisupervised Situation

The method proposed in this article is specifically designed for supervised MS-HeTL, but it can be extended to handle semisupervised classification tasks. That is to say, in addition to using limited labeled samples in the TD, unlabeled samples

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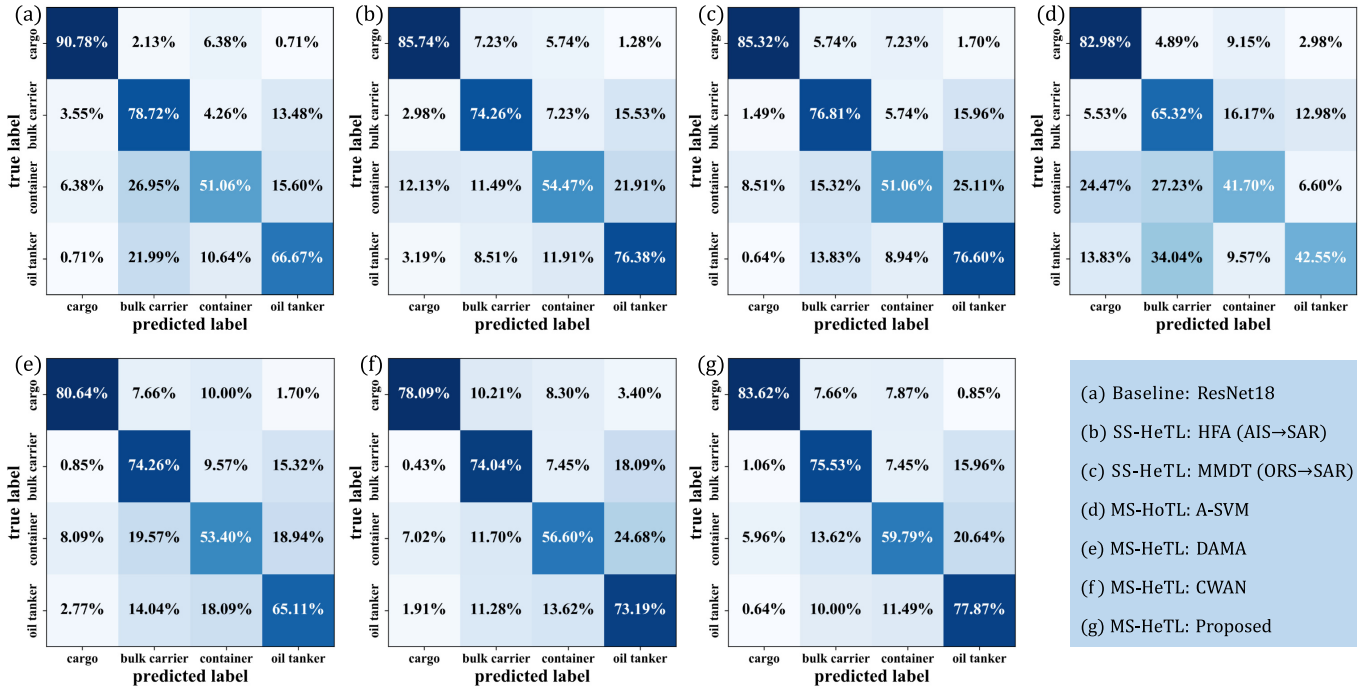


Fig. 3. Confusion matrices of different methods on the FUSAR dataset, where "→" denotes the transfer direction. (a) Baseline: ResNet18. (b) SS-HeTL: HFA (AIS→SAR). (c) SS-HeTL: MMDT (ORS→SAR). (d) MS-HoTL: A-SVM. (e) MS-HeTL: DAMA. (f) MS-HeTL: CWAN. (g) MS-HeTL: proposed.

TABLE IV
CLASS EQUILIBRIUM ANALYSIS

Category	Method	Mean±Std
Baseline	ResNet18	71.81 ± 16.98
SS-HeTL	HFA (AIS→SAR)	72.71 ± 13.14
	MMDT (ORS→SAR)	72.45 ± 14.83
MS-HoTL	A-SVM	58.14 ± 19.85
MS-HeTL	DAMA	68.35 ± 11.83
	CWAN	70.48 ± 9.50
	Proposed	74.20 ± 10.19

TABLE V
PERFORMANCE OF SEMISUPERVISED METHODS ON HR-SAR AND FUSAR DATASETS. (ACC. ± STD. %)

Category	Methods	SDs	TDs	
			HR-SAR	FUSAR
SS-HeTL	SHFA [39]	AIS	81.35 ± 4.50	63.78 ± 7.57
	TNT [64]		75.29 ± 7.41	51.27 ± 5.03
	STN [65]		81.16 ± 3.94	64.42 ± 5.12
	SSAN [66]		84.79 ± 2.51	70.28 ± 4.35
	SHFA [39]	ORS	82.98 ± 2.86	68.19 ± 4.70
	TNT [64]		82.21 ± 2.56	72.96 ± 4.69
	STN [65]		81.01 ± 3.48	72.33 ± 5.04
	SSAN [66]		84.72 ± 2.48	73.05 ± 3.84
MS-HeTL	CWAN	AIS & ORS	82.70 ± 2.72	70.88 ± 5.00
	Proposed$_{semi}$		**86.88 ± 1.93**	**74.10 ± 2.82**

can also be used to assist classifier learning. Suppose that there are n_u unlabeled instances $\{\mathbf{x}_u^j \in \mathbb{R}^{d_t}\}_{j=1}^{n_u}$; we can utilize SVM$_t$ to assign pseudolabel for $\mathbf{x}_u^j$ and then consider it with labeled instances from SDs together and incorporate into the SVM training framework to learn the label prediction function. For the detailed incorporating process, please refer to Appendix B.

As a supplementary experiment, we compared the proposed semisupervised solution (termed proposed$_{\text{semi}}$) with state-of-the-art semisupervised SS-HeTL and MS-HeTL methods and reported the results in Table V. By comparing the performance of semi-supervised heterogeneous feature augmentation (SHFA), transfer neural trees (TNT), and soft transfer network (STN) in Table V with the SS-HeTL methods listed in Table III, it can be seen that, although unlabeled samples are introduced to assist target classifier learning, it is not enough to ensure that semisupervised methods are bound to get better performance than supervised methods (considering they only use labeled samples). The quality of the semi-supervised methods has a lot to do with the processing of pseudolabels. Since simultaneous semantic alignment network (SSAN) can effectively reduce the negative effects caused by falsely assigned pseudolabel, its performance has been improved to a certain extent compared to the supervised SS-HeTL methods in Table III. The proposed$_{\text{semi}}$ outperforms SSAN on the HR-SAR dataset by about 2.0% and on the FUSAR dataset by about 4.0% (AIS as SD) and 1.0% (ORS as SD), respectively. Compared to CWAN, which is a specialized semisupervised MS-HeTL method, the proposed$_{\text{semi}}$ shows even more advantages; it is higher than CWAN by 4.18% and 3.22% on two datasets, respectively.

V. CONCLUSION AND FUTURE WORK

In recent years, research on ship classification in SAR imagery has got more and more attention. Research on improving task performance in the case where there are only limited labeled instances in the TD (i.e., SAR imagery domain) through knowledge transfer from other SD(s) (such as AIS or

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ORS) is in the ascendant. However, the existing studies only solve the problem of HoTL from a single SD (SS-HoTL). This study takes a step forward and attempts to address a more meaningful and challenging problem, that is, transferring knowledge from both AIS and ORS domains to explore and exploit complementary knowledge cross these two SDs for the purpose of assisting ship classification in SAR domain. More importantly, this study develops an MS-HeTL technique that liberates the restriction of utilizing the exact same features for all domains, allowing each domain represented by more appropriate features and, thus, more likely to improve the ship classification performance even further. Extensive experiments on two SAR datasets, HR-SAR and FUSAR, show that the proposed methods can fully exploit the complementarity of the AIS domain and the ORS domain and reasonably balance the weight between domains and, thereby, significantly improve task performance on the target SAR domain.

Improving ship classification performance in the SAR domain based on TL (single-source/multisource, homogeneous/heterogeneous, and machine learning/deep learning) is still a vast blue ocean to be developed, and there are still many issues that need to be explored in depth. Some open questions left from our research in this article are worth solving in future work. First, considering that ORS can provide more discriminative features than AIS, we originally expected that the transfer from ORS to SAR would bring a more significant improvement than that from AIS to SAR, but the result was not as good as we expected. Is it a data problem, a feature problem, or a method problem? It is a topic that needs further research. Second, the proposed method performs feature extraction, TL, and classifier learning step by step, so it is difficult to obtain a global optimal solution. In future work, we plan to use deep TL to directly process single or multiple image domains (ORS, SAR, and so on) while embedding the AIS domain (whose feature cannot be directly extracted with convolutional neural networks) into the deep neural network in an appropriate way, so as to achieve end-to-end optimization for the purpose of achieving the global optimal solution.

APPENDIX A
INCORPORATING MS-HFA WITH SVM TRAINING FRAMEWORK

We first declare some necessary notation representations here; we let superscript $(\cdot)^T$ denote the transpose operation and let trace$(\cdot)$ denote the trace operation. We let $\mathbf{O}_{p\times q}$ as the $p \times q$ all zeroes matrix and $\mathbf{E}_p$ as the $p \times p$ identity matrix. We also let $\mathbf{0}_p$, $\mathbf{1}_p$ as the $p \times 1$ column vector of all zeroes and all ones, and let $\|\cdot\|_2$, $\|\cdot\|_F$ as the Euclidean norm and Frobenius norm. $\mathbf{a} \circ \mathbf{b}$ means Hadamard product between the vectors $\mathbf{a}$ and $\mathbf{b}$. $\mathbf{a} \le \mathbf{b}$ means the inequality that $a_i \le b_i$, where $a_i \in \mathbf{a}$ and $b_i \in \mathbf{b}$. In addition, we define $\mathbf{X}_{si} = [\mathbf{x}_{si}^1, \ldots, \mathbf{x}_{si}^{n_{si}}] \in \mathbb{R}^{d_{si}\times n_{si}}$ and $\mathbf{X}_t = [\mathbf{x}_t^1, \ldots, \mathbf{x}_t^{n_t}] \in \mathbb{R}^{d_t\times n_t}$ as the feature matrices for labeled instances in SD and TD, respectively.

A. Derive to Dual Problem

Based on the defined optimization formulation of the proposed method in (4)–(7), we can derive its dual form by using the Lagrangian multiplier method. In detail, we introduce dual variables $\alpha_{si}^1, \ldots, \alpha_{si}^{n_{si}}$ for instances in $\mathcal{D}_{si}$ and $\alpha_t^1, \ldots, \alpha_t^{n_t}$ for instances in $\mathcal{D}_t$, respectively. Then, we set the derivatives of (4)–(7) with respect to $\mathbf{w}$, b, ξ_{si}^j, and ξ_t^j to zero and obtain the Karush–Kuhn–Tucker (KKT) conditions as

$$\mathbf{w} = \sum_{i=1}^{m}\sum_{j=1}^{n_{si}} \alpha_{si}^j y_{si}^j \varphi_{si}\left(\mathbf{x}_{si}^j\right) + \sum_{j=1}^{n_t} \alpha_t^j y_t^j \varphi_t\left(\mathbf{x}_t^j\right) \tag{9}$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n_{si}} \alpha_{si}^j y_{si}^j + \sum_{j=1}^{n_t} \alpha_t^j y_t^j = 0 \tag{10}$$

$$\left\{0 \le \alpha_{si}^1, \ldots, \alpha_{si}^{n_{si}} \le C\theta_{si}\right\}_{i=1}^{m} \tag{11}$$

$$0 \le \alpha_t^1, \ldots, \alpha_t^{n_t} \le C\theta_t. \tag{12}$$

Thus, we can arrive the dual problem as follows with KKT conditions:

$$\begin{aligned} &\min_{\mathbf{P}_{si},\mathbf{P}_t} \max_{\boldsymbol{\alpha}} \ \mathbf{1}_n^T\boldsymbol{\alpha} - \frac{1}{2}(\boldsymbol{\alpha}\circ\mathbf{y})^T\mathbf{G}_{\mathbf{P}_{si},\mathbf{P}_t}(\boldsymbol{\alpha}\circ\mathbf{y}) \\ &\text{s.t. } \mathbf{y}^T\boldsymbol{\alpha} = 0, \quad \mathbf{0}_n \le \boldsymbol{\alpha} \le \mathbf{C}_n \\ &\qquad \left\{\|\mathbf{P}_{si}\|_F^2 \le \lambda_{si}\right\}_{i=1}^{m}, \quad \|\mathbf{P}_t\|_F^2 \le \lambda_t \end{aligned} \tag{13}$$

where $n = n_t + \sum_{i=1}^{m} n_{si}$ means the number of total training instances from all domains. The vector $\boldsymbol{\alpha} = [\alpha_{s1}^1, \ldots, \alpha_{s1}^{n_{s1}}, \ldots, \alpha_{sm}^1, \ldots, \alpha_{sm}^{n_{sm}}, \alpha_t^1, \ldots, \alpha_t^{n_t}]^T \in \mathbb{R}^n$ is composed by all n dual variables. $\mathbf{y} = [y_{s1}^1, \ldots, y_{s1}^{n_{s1}}, \ldots, y_{sm}^1, \ldots, y_{sm}^{n_{sm}}, y_t^1, \ldots, y_t^{n_t}]^T \in \mathbb{R}^n$ is composed by the label corresponding to instances, and note that $y_{si}^j, y_t^j \in \{1, -1\}$ are the prebinarized labels of the training instances. $\mathbf{G}_{\mathbf{P}_{si},\mathbf{P}_t} = \mathbf{U}^T\mathbf{U} \in \mathbb{R}^{n\times n}$ is the derived kernel matrix for all domains with

$$\mathbf{U} = \begin{bmatrix} \mathbf{P}_{s1}\mathbf{X}_{s1} & \mathbf{P}_{s2}\mathbf{X}_{s2} & \cdots & \mathbf{P}_{sm}\mathbf{X}_{sm} & \mathbf{P}_t\mathbf{X}_t \\ \mathbf{X}_{s1} & \mathbf{O}_{d_{s1}\times n_{s2}} & \cdots & \mathbf{O}_{d_{s1}\times n_{sm}} & \mathbf{O}_{d_{s1}\times n_t} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{O}_{d_{sm}\times n_{s1}} & \mathbf{O}_{d_{sm}\times n_{s2}} & \cdots & \mathbf{X}_{sm} & \mathbf{O}_{d_{sm}\times n_t} \\ \mathbf{O}_{d_t\times n_{s1}} & \mathbf{O}_{d_t\times n_{s2}} & \cdots & \mathbf{O}_{d_t\times n_{sm}} & \mathbf{X}_t \end{bmatrix}. \tag{14}$$

Moreover, we define a constant vector $\mathbf{C}_n$ as the constraint to $\boldsymbol{\alpha}$, i.e., $\mathbf{C}_n = C[\theta_{s1}\mathbf{1}_{n_{s1}}^T, \ldots, \theta_{sm}\mathbf{1}_{n_{sm}}^T, \theta_t\mathbf{1}_{n_t}^T]^T \in \mathbb{R}^n$.

Generally speaking, to solve the dual problem in (13), the dimension of the common subspace (i.e., d_c) must be given in advance. However, it is usually nontrivial to determine the optimal value of d_c. Moreover, we need to solve each feature projection, respectively, which is complex and inefficient. Inspired by [39] and [42], we simplify the dual problem (13) by introduce $\mathbf{H} = [\mathbf{P}_{s1}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t]^T[\mathbf{P}_{s1}, \ldots, \mathbf{P}_{sm}, \mathbf{P}_t] \in \mathbb{R}^{d\times d}$ as an intermediate matrix, where $d = d_t + \sum_{i=1}^{m} d_{si}$ is the sum of feature dimension of all domains. $\mathbf{H}$ is a PSD matrix, i.e., $\mathbf{H} \succeq 0$. By introducing $\mathbf{H}$, the parameter d_c can be thrown away, and the common subspace becomes latent because we do not need to solve $\mathbf{P}_{si}$ and $\mathbf{P}_t$ accurately. Therefore, taking the $\mathbf{H}$ into consideration, we can receive the simplified dual

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form as follows:

$$\begin{aligned}\min_{\mathbf{H}\succeq 0}\ \max_{\boldsymbol{\alpha}}\ & \mathbf{1}_n^T\boldsymbol{\alpha}-\frac{1}{2}(\boldsymbol{\alpha}\circ\mathbf{y})^T\mathbf{G}_{\mathbf{H}}(\boldsymbol{\alpha}\circ\mathbf{y})\\ \text{s.t. }& \mathbf{y}^T\boldsymbol{\alpha}=0,\ \mathbf{0}_n\le\boldsymbol{\alpha}\le\mathbf{C}_n\\ & \text{trace}(\mathbf{H})\le\lambda\end{aligned}\tag{15}$$

where $\mathbf{G}_{\mathbf{H}}=\mathbf{V}^T(\mathbf{H}+\mathbf{E}_d)\mathbf{V}$, $\mathbf{V}\in\mathbb{R}^{d\times n}$ and

$$\mathbf{V}=\begin{bmatrix}\mathbf{X}_{s1} & \mathbf{O}_{d_{s1}\times n_{s2}} & \cdots & \mathbf{O}_{d_{s1}\times n_{sm}} & \mathbf{O}_{d_{s1}\times n_t}\\ \mathbf{O}_{d_{s2}\times n_{s1}} & \mathbf{X}_{s2} & \cdots & \mathbf{O}_{d_{s2}\times n_{sm}} & \mathbf{O}_{d_{s2}\times n_t}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \mathbf{O}_{d_{sm}\times n_{s1}} & \mathbf{O}_{d_{sm}\times n_{s2}} & \cdots & \mathbf{X}_{sm} & \mathbf{O}_{d_{sm}\times n_t}\\ \mathbf{O}_{d_t\times n_{s1}} & \mathbf{O}_{d_t\times n_{s2}} & \cdots & \mathbf{O}_{d_t\times n_{sm}} & \mathbf{X}_t\end{bmatrix}.\tag{16}$$

In addition, $\mathbf{E}_d\in\mathbb{R}^{d\times d}$ is an identity matrix, and $\lambda=\lambda_t+\sum_{i=1}^m\lambda_{si}$.

B. Generalize to Nonlinear Transformation

As the transformation metric $\mathbf{H}$ is based on linear feature transformation, the solve process is computationally expensive when feature with very high dimension. Therefore, we utilize the kernelization strategy to make $\mathbf{H}$ be efficiently solved in the kernel space, resulting in nonlinear transformation, which can make the computational complexity independent of feature dimensions (only dependent on the number of controllable training instances).

Specifically, we define the nonlinear feature transformation function for $\mathcal{D}_{si}$ as $\psi_{si}(\cdot)$; thus, the new feature matrix of $\mathcal{D}_{si}$ is represented as $\Psi_{si}=[\psi_{si}(\mathbf{x}_{si}^1),\ldots,\psi_{si}(\mathbf{x}_{si}^{n_{si}})]\in\mathbb{R}^{\widehat{d}_{si}\times n_{si}}$, and the corresponding kernel is $\mathbf{K}_{si}=\Psi_{si}^T\Psi_{si}\in\mathbb{R}^{n_{si}\times n_{si}}$, where $\widehat{d}_{si}$ means the new feature dimension of SDs instance after nonlinear feature transformation $\psi_{si}(\cdot)$. Analogously, we define the nonlinear feature transformation function for $\mathcal{D}_t$ as $\psi_t(\cdot)$, so the new feature matrix of TD is represented as $\Psi_t=[\psi_t(\mathbf{x}_t^1),\ldots,\psi_t(\mathbf{x}_t^{n_t})]\in\mathbb{R}^{\widehat{d}_t\times n_t}$, and the corresponding kernel is $\mathbf{K}_t=\Psi_t^T\Psi_t\in\mathbb{R}^{n_t\times n_t}$, where $\widehat{d}_t$ means the new feature dimension of TD instance after the nonlinear feature transformation.

Correspondingly, the derived kernel $\mathbf{G}_{\mathbf{H}}$ in (15) can be rewritten as $\mathbf{G}_{\mathbf{H}}=\Psi^T(\mathbf{H}+\mathbf{E}_{\widehat{d}})\Psi$, where $\widehat{d}=\widehat{d}_t+\sum_{i=1}^m\widehat{d}_{si}$, $\Psi\in\mathbb{R}^{\widehat{d}\times n}$, and

$$\Psi=\begin{bmatrix}\Psi_{s1} & \mathbf{O}_{\widehat{d}_{s1}\times n_{s2}} & \cdots & \mathbf{O}_{\widehat{d}_{s1}\times n_{sm}} & \mathbf{O}_{\widehat{d}_{s1}\times n_t}\\ \mathbf{O}_{\widehat{d}_2\times n_{s1}} & \Psi_{s2} & \cdots & \mathbf{O}_{\widehat{d}_{s2}\times n_{sm}} & \mathbf{O}_{\widehat{d}_{s2}\times n_t}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \mathbf{O}_{\widehat{d}_{sm}\times n_{s1}} & \mathbf{O}_{\widehat{d}_{sm}\times n_{s2}} & \cdots & \Psi_{sm} & \mathbf{O}_{\widehat{d}_{sm}\times n_t}\\ \mathbf{O}_{\widehat{d}_t\times n_{s1}} & \mathbf{O}_{\widehat{d}_t\times n_{s2}} & \cdots & \mathbf{O}_{\widehat{d}_t\times n_{sm}} & \Psi_t\end{bmatrix}.\tag{17}$$

However, the dimension of $\mathbf{H}$ is still cannot be determined because the forms of $\psi_{si}(\cdot)$ and $\psi_t(\cdot)$ are usually unexplicit. Inspired by [44], we define a nonlinear transformation matrix $\widehat{\mathbf{N}}\in\mathbb{R}^{n\times n}$ to satisfy $\mathbf{H}=\Psi\mathbf{K}^{-(1/2)}\widehat{\mathbf{N}}\mathbf{K}^{-(1/2)}\Psi^T$ [44], where $\mathbf{K}^{-(1/2)}$ means the inverse of the symmetric square root of kernel matrix $\mathbf{K}$, and

$$\mathbf{K}=\begin{bmatrix}\mathbf{K}_{s1} & \mathbf{O}_{n_{s1}\times n_{s2}} & \cdots & \mathbf{O}_{n_{s1}\times n_{sm}} & \mathbf{O}_{n_{s1}\times n_t}\\ \mathbf{O}_{n_{s2}\times n_{s1}} & \mathbf{K}_{s2} & \cdots & \mathbf{O}_{n_{s2}\times n_{sm}} & \mathbf{O}_{n_{s2}\times n_t}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \mathbf{O}_{n_{sm}\times n_{s1}} & \mathbf{O}_{n_{sm}\times n_{s2}} & \cdots & \mathbf{K}_{sm} & \mathbf{O}_{n_{sm}\times n_t}\\ \mathbf{O}_{n_t\times n_{s1}} & \mathbf{O}_{n_t\times n_{s2}} & \cdots & \mathbf{O}_{n_t\times n_{sm}} & \mathbf{K}_t\end{bmatrix}.\tag{18}$$

Thus, $\text{trace}(\widehat{\mathbf{N}})=\text{trace}(\mathbf{H})\le\lambda$, and $\mathbf{G}_{\mathbf{H}}=\Psi^T(\mathbf{H}+\mathbf{E}_{\widehat{d}})\Psi=\mathbf{K}^{(1/2)}(\widehat{\mathbf{N}}+\mathbf{E}_n)\mathbf{K}^{(1/2)}=\mathbf{G}_{\widehat{\mathbf{N}}}$. Hence, the dual problem in nonlinear situation can be written as follows:

$$\begin{aligned}\min_{\widehat{\mathbf{N}}\succeq 0}\ \max_{\boldsymbol{\alpha}}\ & \mathbf{1}_n^T\boldsymbol{\alpha}-\frac{1}{2}(\boldsymbol{\alpha}\circ\mathbf{y})^T\mathbf{G}_{\widehat{\mathbf{N}}}(\boldsymbol{\alpha}\circ\mathbf{y})\\ \text{s.t. }& \mathbf{y}^T\boldsymbol{\alpha}=0,\quad \mathbf{0}_n\le\boldsymbol{\alpha}\le\mathbf{C}_n\\ & \text{trace}\left(\widehat{\mathbf{N}}\right)\le\lambda.\end{aligned}\tag{19}$$

Therefore, we can solve $\widehat{\mathbf{N}}$ in (19) rather than directly solving $\mathbf{H}$. Noting that the size of $\widehat{\mathbf{N}}$ is independent of the new feature dimensions $\widehat{d}_{si}$ and $\widehat{d}_t$ obtained by nonlinear feature transformations.

C. Solution

To solve the optimal problem in (19), we reformulate it as a convex MKL problem and use the MKL solver [45], [67] to obtain the global solution. Specifically, we decompose $\widehat{\mathbf{N}}$ into a linear combination of a series of rank-one normalized PSD matrices, i.e., $\widehat{\mathbf{N}}=\sum_{r=1}^{\infty}\eta_r\mathbf{Z}_r$, where $\eta_r\ge 0$. We define $\mathcal{Z}=\{\mathbf{Z}_r\}_{r=1}^{\infty}$ as the set of $\mathbf{Z}_r$ and the coefficient vector as $\boldsymbol{\eta}=[\eta_1,\ldots,\eta_\infty]^T$. Thus, the solution process of $\widehat{\mathbf{N}}$ is transformed into solving $\mathbf{Z}_r$ and the corresponding coefficients η_r, the $\text{trace}(\widehat{\mathbf{N}})=\text{trace}(\sum_{r=1}^{\infty}\eta_r\mathbf{Z}_r)=\sum_{r=1}^{\infty}\eta_r\text{trace}(\mathbf{Z}_r)=\mathbf{1}_\infty^T\boldsymbol{\eta}$. By setting $\boldsymbol{\eta}\leftarrow(1/\lambda)\boldsymbol{\eta}$, we reformulate the optimization problem (19) as

$$\begin{aligned}\min_{\boldsymbol{\eta}}\ \max_{\boldsymbol{\alpha}}\ & \mathbf{1}_n^T\boldsymbol{\alpha}-\frac{1}{2}(\boldsymbol{\alpha}\circ\mathbf{y})^T\sum_{r=1}^{\infty}\eta_r\mathbf{M}_r(\boldsymbol{\alpha}\circ\mathbf{y})\\ \text{s.t. }& \mathbf{y}^T\boldsymbol{\alpha}=0,\quad \mathbf{0}_n\le\boldsymbol{\alpha}\le\mathbf{C}_n\\ & \mathbf{1}_\infty^T\boldsymbol{\eta}\le 1,\quad \mathbf{0}_\infty\le\boldsymbol{\eta}\end{aligned}\tag{20}$$

where $\mathbf{M}_r=\mathbf{K}^{(1/2)}(\lambda\mathbf{Z}_r+\mathbf{E}_n)\mathbf{K}^{(1/2)}$. It is an infinite kernel learning problem [67], and each base kernel is $\mathbf{M}_r$, which can be solved by using existed MKL solver [45].

Although there are an infinite number of matrices in $\mathcal{Z}$, a finite number of rank-one matrices are sufficiently effective to represent the ideal $\widehat{\mathbf{N}}$ [46]. Thus, we can approximate (20) by using the cutting plane technique proposed in [67]. In this way, each base kernel $\mathbf{M}_r$ can be obtained by iterative selection as follows:

$$\begin{aligned}\max_{\mathbf{Z}\in\mathcal{Z}}\ & \frac{1}{2}(\boldsymbol{\alpha}\circ\mathbf{y})^T\mathbf{G}_{\mathbf{Z}}(\boldsymbol{\alpha}\circ\mathbf{y})\\ \text{s.t. }& \mathbf{y}^T\boldsymbol{\alpha}=0,\quad \mathbf{0}_n\le\boldsymbol{\alpha}\le\mathbf{C}_n\end{aligned}\tag{21}$$

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where $\mathbf{G_Z} = \mathbf{K}^{(1/2)}(\lambda\mathbf{Z} + \mathbf{E}_n)\mathbf{K}^{(1/2)}$, and it has a closed form solution as $\mathbf{Z} = \mathbf{z}\mathbf{z}^T \in \mathbb{R}^{n\times n}$ with $\mathbf{z} = (\mathbf{K}^{(1/2)}(\boldsymbol{\alpha} \circ \mathbf{y})/\|\mathbf{K}^{(1/2)}(\boldsymbol{\alpha} \circ \mathbf{y})\|_2)$.

The solution steps can be stated as follows. First, we set $r = 1$ and initialize the set $\mathcal{Z}_1$ with $\mathbf{Z}_1 = \mathbf{z}_1\mathbf{z}_1^T$ and $\mathbf{z}_1 = (\mathbf{1}_n/(n)^{1/2})$. Then, in following a repeat procedure, in the rth repeat, we can solve the optimal $\boldsymbol{\eta}$ and $\boldsymbol{\alpha}$ in (20) by an MKL solver based on the current set $\mathcal{Z}_r$. After that, we obtain a new rank-one PSD matrix $\mathbf{Z}_{r+1}$ based on current $\boldsymbol{\eta}$ and $\boldsymbol{\alpha}$ by (21). Then, we add $\mathbf{Z}_{r+1}$ into $\mathcal{Z}_r$ and to obtain a new union set as $\mathcal{Z}_{r+1} = \mathcal{Z}_r \cup \mathbf{Z}_{r+1}$, and then, let $\mathcal{Z}_{r+1}$ be the current set and be used to solve the MKL problem in (20) again. The above steps are repeated until convergence. Finally, we can output the global optimal solution $\boldsymbol{\eta}^*$ and $\boldsymbol{\alpha}^*$ according to the theorem proved in [39], as well as the final set $\mathcal{Z}^*$, which is a set of all $\mathbf{Z}_r$. Therefore, we can obtain $\widehat{\mathbf{N}} = \lambda\sum_r \eta_r\mathbf{Z}_r$, where η_r and $\mathbf{Z}_r$ represent the rth element in the vector $\boldsymbol{\eta}^*$ and the set $\mathcal{Z}^*$, respectively. Then, we can obtain the target decision function for any unlabeled TD instance $\mathbf{x}$ as

$$f(\mathbf{x}) = (\boldsymbol{\alpha} \circ \mathbf{y})^T\mathbf{K}^{\frac{1}{2}}\left(\widehat{\mathbf{N}} + \mathbf{E}_n\right)\begin{bmatrix}\mathbf{O}_{(n-n_t)\times n_t}\\ \mathbf{K}_t^{-\frac{1}{2}}\end{bmatrix}\mathbf{k}_t + b \quad (22)$$

where $\mathbf{k}_t = [k(\mathbf{x}_t^1, \mathbf{x}), \ldots, k(\mathbf{x}_t^{n_t}, \mathbf{x})]^T$ and $k(\cdot,\cdot)$ is a predefined kernel function for two instances from TD.

Appendix B
Extending to Semisupervised Situation

In this part, we further propose the semisupervised variant of the proposed method, which aims at performing TL by using both the labeled and unlabeled TD instances.

A. Define Optimization Formulation

Similar to SHFA [39], we also define the optimization formulation of the semisupervised method variant based on the ρ-SVM with squared hinge loss as follows:

$$\min_{\mathbf{P}_{si},\mathbf{P}_t}\ \min_{\mathbf{w},b,\rho,\xi_{si}^j,\xi_t^j,\xi_u^j}\ \frac{1}{2}\mathbf{w}^T\mathbf{w} + \frac{1}{2}b^2 - \rho + \frac{C}{2}\left(\sum_{i=1}^{m}\theta_{si}\sum_{j=1}^{n_{si}}\left(\xi_{si}^j\right)^2 + \theta_t\sum_{j=1}^{n_t}\left(\xi_t^j\right)^2\right) + \frac{C_u}{2}\theta_t\sum_{j=1}^{n_u}\left(\xi_u^j\right)^2 \quad (23)$$

$$\text{s.t. } \left\{y_{si}^j\left(\mathbf{w}^T\varphi_{si}\left(\mathbf{x}_{si}^j\right) + b\right) \geq \rho - \xi_{si}^j, \xi_{si}^j \geq 0\right\}_{i=1}^{m} \quad (24)$$

$$y_t^j\left(\mathbf{w}^T\varphi_t\left(\mathbf{x}_t^j\right) + b\right) \geq \rho - \xi_t^j, \quad \xi_t^j \geq 0 \quad (25)$$

$$\hat{y}_u^j\left(\mathbf{w}^T\varphi_t\left(\mathbf{x}_u^j\right) + b\right) \geq \rho - \xi_u^j, \quad \xi_u^j \geq 0 \quad (26)$$

$$\left\{\|\mathbf{P}_{si}\|_F^2 \leq \lambda_{si}\right\}_{i=1}^{m}, \quad \|\mathbf{P}_t\|_F^2 \leq \lambda_t \quad (27)$$

where φ_{si} and φ_t are the feature augmentation transformations, as defined in (2) and (3), respectively. C_u is the penalty coefficient for unlabeled TD instances. ξ_u^j and $\hat{y}_u^j$ are the slack variables and pseudolabels of $\mathbf{x}_u^j$.

B. Nonlinear Feature Transformation

For simply, we mainly show the generalized case with nonlinear feature transformation. We first redefine the kernel matrix $\mathbf{K}$ in (18) as

$$\mathbf{K} = \begin{bmatrix}\mathbf{K}_{s1} & \mathbf{O}_{n_{s1}\times n_{s2}} & \cdots & \mathbf{O}_{n_{s1}\times n_{sm}} & \mathbf{O}_{n_{s1}\times n_{tu}}\\ \mathbf{O}_{n_{s2}\times n_{s1}} & \mathbf{K}_{s2} & \cdots & \mathbf{O}_{n_{s2}\times n_{sm}} & \mathbf{O}_{n_{s2}\times n_{tu}}\\ \vdots & \vdots & \ddots & \vdots & \vdots\\ \mathbf{O}_{n_{sm}\times n_{s1}} & \mathbf{O}_{n_{sm}\times n_{s2}} & \cdots & \mathbf{K}_{sm} & \mathbf{O}_{n_{sm}\times n_{tu}}\\ \mathbf{O}_{n_t\times n_{s1}} & \mathbf{O}_{n_t\times n_{s2}} & \cdots & \mathbf{O}_{n_t\times n_{sm}} & \mathbf{K}_{tu}\end{bmatrix} \quad (28)$$

where $n_{tu} = n_t + n_u$ means the number of labeled and unlabeled instances in TD and $\mathbf{K}_{tu} \in \mathbb{R}^{n_{\text{tu}}\times n_{\text{tu}}}$ is the kernel of all labeled and unlabeled TD instances. Then, by redefining the nonlinear transformation $\widehat{\mathbf{N}} \in \mathbb{R}^{(n+n_u)\times(n+n_u)}$, we can derive the simplified dual form of (23) in nonlinear case as follows:

$$\begin{aligned}\min_{\widehat{\mathbf{N}}\succeq 0,\mathbf{Y}}\ \max_{\boldsymbol{\alpha}}\ & -\frac{1}{2}\mathcal{A}^T\left(\mathbf{Q}_{\widehat{\mathbf{N}},\mathbf{Y}} + \mathbf{B}\right)\mathcal{A}\\ \text{s.t. } & \mathbf{0} \leq \mathcal{A}, \quad \mathbf{1}^T\mathcal{A} = 1\\ & \text{trace}\left(\widehat{\mathbf{N}}\right) \leq \lambda\end{aligned} \quad (29)$$

where $\mathbf{Q}_{\widehat{\mathbf{N}},\mathbf{Y}} = (\mathbf{K}^{(1/2)}(\widehat{\mathbf{N}} + \mathbf{E}_{n+n_u})\mathbf{K}^{(1/2)} + \mathbf{1}\mathbf{1}^T) \circ (\mathbf{Y}\mathbf{Y}^T) \in \mathbb{R}^{(n+n_u)\times(n+n_u)}$, and $\mathbf{Y} = [\mathbf{y}^T, \hat{\mathbf{y}}_u^T]^T$ is the label vector of all labeled and unlabeled instances. $\mathcal{A}$ is the vector composed by all dual variables for all labeled and unlabeled instances, i.e., $\mathcal{A} = [\alpha_{s1}^1, \ldots, \alpha_{s1}^{n_{s1}}, \ldots, \alpha_{sm}^1, \ldots, \alpha_{sm}^{n_{sm}}, \alpha_t^1, \ldots, \alpha_t^{n_t}, \alpha_u^1, \ldots, \alpha_u^{n_u}]^T \in \mathbb{R}^{(n+n_u)\times(n+n_u)}$. $\mathbf{B} \in \mathbb{R}^{(n+n_u)\times(n+n_u)}$ is a diagonal matrix with the diagonal elements as $(1/C\theta_{si})$, $(1/C\theta_t)$ for labeled instances from $\mathcal{D}_{si}$ and $\mathcal{D}_t$, and as $(1/C_u\theta_t)$ for the unlabeled TD instances.

C. Solution

We still use the similar MKL strategy as used in MS-HFA to solve the dual problem in (29) by reformulating it as an MKL problem. Specifically, we first decompose $\widehat{\mathbf{N}}$ into a linear combination of a series of rank-one PSD matrices, i.e., $\widehat{\mathbf{N}} = \sum_{r=1}^{\infty}\eta_r\mathbf{Z}_r$, where $\eta_r \geq 0$ is the coefficient combination coefficients of $\mathbf{Z}_r$. Thus, the solution process of $\widehat{\mathbf{N}}$ is transformed into solving $\mathbf{Z}_r$ and the corresponding coefficients η_r, and the readers can refer [39] for more details. The optimization algorithm can finally converge to the global optimal solution according to the theoretical proof in [39]. Based on the optimal solution obtained in the rth iterations, we can predict the label of TD instance $\mathbf{x}$ by using the following function:

$$f(\mathbf{x}) = \sum_r \mu_r(\mathcal{A} \circ \mathbf{Y}_r)^T\mathbf{K}^{\frac{1}{2}}(\lambda\mathbf{Z}_r + \mathbf{E})\begin{bmatrix}\mathbf{O}_{(n-n_t)\times n_{tu}}\\ \mathbf{K}_t^{-\frac{1}{2}}\end{bmatrix}\mathbf{k}_{tu} + b \quad (30)$$

where μ_r denotes the learned relaxation variable from the MKL solver in the rth step iteration [39] and $\mathbf{Y}_r$ denotes the updated label vector in the rth step iteration. $\mathbf{k}_{tu} = [k(\mathbf{x}_t^1, \mathbf{x}), \ldots, k(\mathbf{x}_t^{n_t}, \mathbf{x}), k(\mathbf{x}_u^1, \mathbf{x}), \ldots, k(\mathbf{x}_u^{n_u}, \mathbf{x})]^T$, and $k(\cdot,\cdot)$ is a predefined kernel function for two instances from TD.

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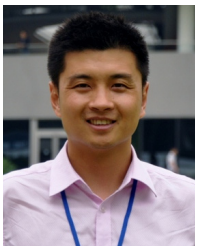
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